CTE

1.Find the highest salaried employee of finance department.

```
-- 1.Find the highest salaried employee of finance department.
-- creates a temporal table that geta all finance employees with their salary
WITH finance_emp AS (
    SELECT e.emp_no, e.first_name, e.last_name, s.salary
    FROM employees e
    -- join to know which department the employee belongs to
    JOIN dept_emp de ON e.emp_no = de.emp_no
    -- join to get the department name
    JOIN departments d ON de.dept_no = d.dept_no
    -- Join to get salary details of the employee
    JOIN salaries s ON e.emp_no = s.emp_no
    -- takes only Finance department employees
    WHERE d.dept_name = "Finance"
)
-- slecting employee details from finance_emp
SELECT emp_no, first_name, last_name, salary
FROM finance_emp
-- returns the highest salary
ORDER BY salary DESC LIMIT 1;
```

esult Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	first_name	last_name	salary
28124	Adhemar	Eiter	134662

2.Current manager of the production department.

```sql
-- 2.Current manager of the production department
-- creates a CTE to get the current manager of each department
WITH current_manager AS(
    select dm.emp_no, dm.dept_no
    FROM dept_manager dm
    -- Current manager has a future to_date
    WHERE dm.to_date = "9999-01-01"
)
--  select details of the manager for the production department
select e.emp_no, e.first_name, e.last_name
FROM current_manager cm JOIN departments d
ON cm.dept_no = d.dept_no
JOIN employees e ON cm.emp_no = e.emp_no
-- filter only the production department
WHERE d.dept_name = "Production";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	first_name	last_name
110420	Oscar	Ghazalie

3.How many managers are worked in the Human Resources department.

```sql
-- 3.How many managers are worked in the Human Resources department
-- creates a CTE to get all managers with their departments
WITH hr_managers AS (
    SELECT dm.emp_no
    FROM dept_manager dm JOIN departments d
    ON dm.dept_no = d.dept_no
    -- filter only Human Resources department
    WHERE d.dept_name = 'Human Resources'
)

-- count of managers works in Human Resources
SELECT COUNT(DISTINCT emp_no) AS total_hr_managers
FROM hr_managers;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

total_hr_managers
2

4.Salary of the Current Manager of the Production Department.

```
-- 4.Salary of the Current Manager of the Production Department

-- select the salary of that current manager
WITH current_manager AS(
    SELECT dm.emp_no
    FROM dept_manager dm JOIN departments d
    ON dm.dept_no = d.dept_no
    WHERE d.dept_name = "production"
    -- current salary has a future to_date
      AND dm.to_date = "9999-01-01"
)

-- select the salary of that current manager
SELECT s.emp_no, s.salary
FROM salaries s
JOIN current_manager cm ON s.emp_no = cm.emp_no
-- current salary has a future to_date
WHERE s.to_date = "9999-01-01";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	salary
110420	56654

5.Find the people whose name starts with a and end with s with only 5 letters , and their salaries.

```
-- 5.Find the people whose name starts with a and end with s with only 5 letters , and their salaries.

-- select employees whose first name
-- starts with 'A', ends with 's', and has only 5 letters
WITH name_filter AS (
    SELECT emp_no, first_name
    FROM employees
    -- A___s add 3 letters between A and S so the total is 5 letters
    WHERE first_name LIKE 'A___s'
)

-- get their current salary
SELECT nf.emp_no, nf.first_name, s.salary
FROM name_filter nf JOIN salaries s
ON nf.emp_no = s.emp_no
-- current salary has a future to_date
WHERE s.to_date = "9999-01-01";
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	salary
110420	56654

6.Find the employees whose hire date and From date are different ,how much difference is there for each one.

```
-- 6.Find the employees whose hire date and From date are different ,how much difference is there for each one.

-- This avoids duplicate emp_no caused by multiple salary records
WITH first_salary AS (
    SELECT emp_no,
        MIN(from_date) AS first_from_date
    FROM salaries
    GROUP BY emp_no
)
-- compare hire date with the first salary start date
SELECT e.emp_no, e.first_name, e.hire_date, fs.first_from_date,
    -- calculate the difference in days between the two dates
    DATEDIFF(fs.first_from_date, e.hire_date) AS difference_in_days
FROM employees e JOIN first_salary fs
ON e.emp_no = fs.emp_no
-- show only employees where hire date and salary start date are different
WHERE e.hire_date <> fs.first_from_date;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

emp_no	first_name	hire_date	first_from_date	difference_in_days
10002	Bezalel	1985-11-21	1996-08-03	3908
10003	Parto	1986-08-28	1995-12-03	3384
10006	Anneke	1989-06-02	1990-08-05	429
10008	Saniya	1994-09-15	1998-03-11	1273
10010	Duangkaew	1989-08-24	1996-11-24	2649
10014	Berni	1987-03-11	1993-12-29	2485
10015	Guoxiang	1987-07-02	1992-09-19	1906
10016	Kazuhito	1995-01-27	1998-02-11	1111
10020	Mayuko	1991-01-26	1997-12-30	2530
10022	Shahaf	1995-08-22	1999-09-03	1473
10023	Bojan	1989-12-17	1999-09-27	3571
10024	Suzette	1997-05-19	1998-06-14	391
10027	Divier	1989-07-07	1995-04-02	2095
10029	Otmar	1985-11-20	1991-09-18	2128
10034	Bader	1988-09-21	1995-04-12	2394
10036	Adamantios	1992-01-03	1992-04-28	116
10045	Moss	1989-09-02	1996-11-16	2632
10052	Heping	1988-05-21	1997-01-31	3177
10053	Sanjiv	1986-02-04	1994-11-13	3204
10054	Mayumi	1995-03-13	1995-07-29	138
10058	Berhard	1987-04-13	1988-04-25	378
10060	Breannda	1987-11-02	1989-05-28	573
10061	Tse	1985-09-17	1989-12-02	1537
10065	Satosi	1988-05-18	1998-05-24	3658
10069	Margareta	1989-11-05	1992-06-14	952
10071	Hisao	1987-10-01	1995-08-05	2865
10072	Hironoby	1988-07-21	1989-05-21	304

7.Highest salaried manager and for which department he is present in, i need dept name ,and manager emp_no as output

```
-- 7.Highest salaried manager and for which department he is present in,
-- i need dept name ,and manager emp_no as output

-- get all current managers with their department and salary
WITH manager_salary AS (
    SELECT dm.emp_no, d.dept_name, s.salary
    FROM dept_manager dm JOIN departments d
    ON dm.dept_no = d.dept_no
    JOIN salaries s ON dm.emp_no = s.emp_no
    -- taking only current managers
    WHERE dm.to_date = '9999-01-01'
      AND s.to_date = '9999-01-01'
)
-- select the manager with the highest salary
SELECT emp_no, dept_name
FROM manager_salary
ORDER BY salary DESC LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	dept_name
110039	Marketing

8.Find which current department manager has less salary than highest salaried employee of that department.

```
-- 8.Find which current department manager has less salary than highest salaried employee of that department

-- get current employees with their department and current salary
WITH current_emp_salary AS (
    SELECT de.dept_no, s.salary
    FROM dept_emp de JOIN salaries s
    ON de.emp_no = s.emp_no
    -- only current employees and their current salaries
    WHERE de.to_date = "9999-01-01"
      AND s.to_date = "9999-01-01"
),
-- find the highest salary in each department
dept_max_salary AS (
    SELECT dept_no, MAX(salary) AS max_salary
    FROM current_emp_salary
    GROUP BY dept_no
),
```

```sql
-- get current managers with their salary and department
current_managers AS (
    SELECT dm.emp_no, dm.dept_no, s.salary
    FROM dept_manager dm JOIN salaries s
    ON dm.emp_no = s.emp_no
    -- only current managers and current salaries
    WHERE dm.to_date = "9999-01-01"
      AND s.to_date = "9999-01-01"
)

-- compares manager salary with highest salary in department
SELECT
    cm.emp_no AS manager_emp_no,
    d.dept_name,
    cm.salary AS manager_salary,
    dms.max_salary AS highest_employee_salary
FROM current_managers cm
JOIN dept_max_salary dms ON cm.dept_no = dms.dept_no
JOIN departments d ON cm.dept_no = d.dept_no
--  manager who has less salary than highest salaried employee of that department
WHERE cm.salary < dms.max_salary;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

manager_emp_no	dept_name	manager_salary	highest_employee_salary
110039	Marketing	106491	143644
110114	Finance	83457	134662
110228	Human Resources	65400	123674
110420	Production	56654	132552
110567	Development	74510	144434
110854	Quality Management	72876	121899
111133	Sales	101987	158220
111534	Research	79393	124181
111939	Customer Service	58745	144866

9.Which department manager has the highest salary among all departments. output : emp_no ,salary,dept_name

```
-- 9.Which department manager has the highest salary among all departments. output : emp_no ,salary,dept_name

-- get all current managers with their department and salary
WITH manager_salary AS (
    SELECT dm.emp_no, d.dept_name, s.salary
    FROM dept_manager dm JOIN departments d
    ON dm.dept_no = d.dept_no
    JOIN salaries s ON dm.emp_no = s.emp_no
    -- only current managers and their current salaries
    WHERE dm.to_date = "9999-01-01"
      AND s.to_date = "9999-01-01"
)
-- select the manager with the highest salary
SELECT emp_no, salary, dept_name
FROM manager_salary
ORDER BY salary DESC LIMIT 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

emp_no	salary	dept_name
110039	106491	Marketing